

Nexivarin 10 mg Tablets — eCTD Submission  
Module 4 — Nonclinical Study Reports

|                 |                 |                 |                              |
|-----------------|-----------------|-----------------|------------------------------|
| Submission Type | NDA — 505(b)(1) | Sequence Number | 0000                         |
| Submission Date | 2021-03-15      | Section         | <a href="#">4.2.1.1</a>      |
| Prepared by     | PharmaCorp Inc. | Confidentiality | Proprietary and Confidential |

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| Section                 | Title                                             | Description                                                                                                                |
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| <a href="#">4.2.1.1</a> | Primary Pharmacodynamics                          | Studies characterizing the pharmacological activity of nexivarin                                                           |
| Study<br>NXV-PD-001     | In Vitro Calcium Channel Inhibition (Patch-Clamp) | IC50 determination for L-type Cav1.2 in VSMCs and cardiac myocytes; GLP compliant                                          |
| Study<br>NXV-PD-002     | In Vivo Antihypertensive Activity (SHR Model)     | 4-week dose-response in spontaneously hypertensive rats ( <a href="#">0.1</a> –10 mg/kg); ED50 ~ <a href="#">0.8</a> mg/kg |

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Section 4.2.1.1  
Primary Pharmacodynamics

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4.2.1.1 Primary Pharmacodynamics — Nexivarin Hydrochloride

Study 4.2.1.1-001: In Vitro Calcium Channel Inhibition (Patch-Clamp)

| Parameter     | Value                                                 |
|---------------|-------------------------------------------------------|
| Study Number  | NXV-PD-001                                            |
| Study Type    | In Vitro Electrophysiology — Whole Cell Patch Clamp   |
| Cell System   | Human coronary artery SMCs & rat ventricular myocytes |
| ICH Guideline | ICH S7A                                               |
| GLP Status    | GLP Compliant (US 21 CFR Part 58)                     |
| Facility      | PharmaCorp Pharmacology Laboratory, Newark, NJ        |
| Study Date    | 2013-09-12 to 2013-11-30                              |

**Objective:** Determine the potency and selectivity of nexivarin hydrochloride for inhibition of L-type (Cav1.2) calcium channels in vascular smooth muscle cells (VSMCs) and cardiac myocytes.

Results:

| Cell Type                 | Channel       | IC50 (nM)  | Hill Coefficient | Notes                          |
|---------------------------|---------------|------------|------------------|--------------------------------|
| Human coronary artery SMC | L-type Cav1.2 | 2.3 ± 0.4  | 1.02             | Concentration-dependent        |
| Rat ventricular myocyte   | L-type Cav1.2 | 18.7 ± 2.1 | 0.98             | ~8-fold lower potency vs. VSMC |
| Human coronary artery SMC | T-type Cav3.1 | > 10,000   | —                | No significant inhibition      |
| Human coronary artery SMC | N-type Cav2.2 | > 10,000   | —                | No significant inhibition      |

**Conclusion:** Nexivarin is a potent, selective L-type calcium channel inhibitor with approximately 8-fold selectivity for vascular smooth muscle over cardiac muscle, consistent with the desired peripheral vasodilatory mechanism of action and a favorable cardiovascular safety profile.

Study 4.2.1.1-002: In Vivo Antihypertensive Activity (SHR Model)

| Parameter        | Value                                                    |
|------------------|----------------------------------------------------------|
| Study Number     | NXV-PD-002                                               |
| Species/Strain   | Spontaneously Hypertensive Rat (SHR), male, 14–16 weeks  |
| Dose Levels      | 0.1, 0.3, 1, 3, 10 mg/kg/day (oral gavage)               |
| Duration         | 4 weeks                                                  |
| Control          | Vehicle (0.5% MC) + normotensive Wistar-Kyoto rats       |
| Primary Endpoint | Systolic blood pressure (SBP, tail-cuff plethysmography) |
| GLP Status       | GLP Compliant                                            |

| Dose (mg/kg)  | Week 0 SBP (mmHg) | Week 4 SBP (mmHg) | ΔSBP (mmHg) | p-value |
|---------------|-------------------|-------------------|-------------|---------|
| Vehicle (SHR) | 188 ± 6           | 192 ± 7           | +4 ± 3      | —       |
| 0.1           | 188 ± 5           | 184 ± 6           | -4 ± 4      | NS      |
| 0.3           | 189 ± 4           | 178 ± 5           | -11 ± 4     | <0.05   |

| Dose (mg/kg)        | Week 0 SBP (mmHg) | Week 4 SBP (mmHg) | ΔSBP (mmHg) | p-value |
|---------------------|-------------------|-------------------|-------------|---------|
| 1.0 (ED50≈0.8)      | 187 ± 6           | 164 ± 5           | -23 ± 5     | <0.001  |
| 3.0                 | 190 ± 5           | 148 ± 6           | -42 ± 5     | <0.001  |
| 10.0                | 188 ± 6           | 135 ± 7           | -53 ± 6     | <0.001  |
| Vehicle (WKY norm.) | 132 ± 4           | 131 ± 4           | -1 ± 2      | —       |

**Conclusion:** Nexivarin dose-dependently reduced SBP in SHR rats with an estimated ED50 of ~0.8 mg/kg/day. Antihypertensive effects were maintained throughout the 4-week treatment period with no tachyphylaxis. No adverse cardiac effects were noted.